
MISALLOCATION AND CAPITAL MARKET INTEGRATION: EVIDENCE FROM INDIA

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ABSTRACT

India liberalized foreign direct investment equity caps across manufacturing industries in a staggered sequence between 1994 and 2006. We estimate the effect of these reforms on capital misallocation—measured as the within-industry dispersion of the marginal revenue product of capital—using a staggered difference-in-differences design on a panel of 51 three-digit NIC industries spanning 1990 to 2020. FDI liberalization reduces MRPK dispersion by 8.7 percent five to nine years after reform and by 10.1 percent beyond ten years. In the first four years post-reform, however, the effect is statistically indistinguishable from zero ($p = 0.60$). The divergence between short-run and long-run effects is decisive ($F = 555.6$, $p < 0.001$). These findings imply that foreign equity liberalization improves allocative efficiency, but only after productive firms have had sufficient time to expand and unproductive firms to exit.

Keywords FDI liberalization; capital misallocation; MRPK dispersion; staggered difference-in-differences; India manufacturing

1 Introduction

Manufacturing firms in developing economies routinely face marginal revenue products of capital (MRPK) that differ by a factor of three or more within the same narrowly defined industry (Hsieh and Klenow, 2009). In India, this dispersion is particularly severe: equalizing marginal products across firms would raise aggregate manufacturing TFP by 40 to 60 percent. Financial frictions—collateral constraints, credit rationing, and limited access to external equity—account for roughly 25 to 50 percent of observed MRPK dispersion (Midrigan and Xu, 2014; Calligaris, Del Gatto, Hassan, Ottaviano, and Schivardi, 2018). India’s phased liberalization of foreign direct investment (FDI) equity caps between 1994 and 2006 offers a quasi-experimental setting to test whether opening the capital account to foreign equity improves the allocation of capital across firms. We ask: did the staggered removal of FDI equity restrictions reduce within-industry capital misallocation in Indian manufacturing, and did any such reduction require time beyond five years to materialize?

We find that FDI liberalization reduces MRPK dispersion, but only at longer horizons. Using a staggered difference-in-differences design on a panel of 51 three-digit NIC industries spanning 1990 to 2020, we estimate that the within-industry standard deviation of log MRPK falls by 8.7 percent five to nine years after liberalization and by 10.1 percent beyond ten years. In the short run—the first four years post-reform—the effect is a precisely estimated zero (coefficient = +0.002, $p = 0.60$). The divergence between short-run and long-run effects is statistically decisive ($F = 555.6$, $p < 0.001$). A 10-percentage-point increase in foreign capital penetration is associated with a 2.7 percent reduction in MRPK dispersion, confirming that the effect operates through the intensity of actual foreign capital exposure rather than merely crossing a *de jure* policy threshold. These estimates are robust to excluding the tariff control variable, ruling out the concern that India’s simultaneous trade liberalization confounds the FDI effect.

This paper makes two contributions. First, we provide the first direct evidence on the causal effect of FDI liberalization on within-industry capital misallocation in a developing economy. Existing work has documented that domestic banking deregulation reduces misallocation in the United States (Bai, Carvalho, and Phillips, 2018) but has also shown that capital inflows can *increase* misallocation when domestic financial intermediation is weak (Gopinath,

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Kalemli-Özcan, Karabarbounis, and Villegas-Sanchez, 2017). Our results resolve this tension for the Indian case: foreign capital does improve allocative efficiency, but the improvement is hidden in short-run analyses that fail to track industries for at least five years after reform. Second, we demonstrate that the timing of the effect matters critically. The standard average treatment effect from a binary pre-post comparison—a 2.1 percent reduction—dramatically understates the long-run benefit by averaging over the null short-run period. Event-time estimation reveals that the economically meaningful effect, an 8 to 10 percent reduction in misallocation, is entirely concentrated at horizons of five years or more.

The paper proceeds as follows. Section 2 situates our analysis in the literatures on capital misallocation, financial liberalization, and Indian firm dynamics. Section 3 describes the industry-year panel, the construction of MRPK dispersion, and the FDI treatment variable. Section 4 develops the staggered difference-in-differences identification strategy and addresses threats to validity. Section 5 presents the main results, event-time dynamics, continuous treatment estimates, and robustness checks. Section 6 concludes with implications for capital account liberalization policy.

2 Literature Review

A large literature following Hsieh and Klenow (2009) measures misallocation as the dispersion of marginal revenue products across firms within an industry. Applying this framework to Indian manufacturing, Hsieh and Klenow estimate that reallocating capital and labor to equalize marginal products would increase aggregate TFP by 40 to 60 percent—the largest estimate among the countries they study. Restuccia and Rogerson (2008) provide the theoretical foundation, showing how idiosyncratic policy distortions to heterogeneous firms reduce aggregate TFP through misallocation. Subsequent work has established financial frictions as a central driver of this dispersion. Midrigan and Xu (2014) show that collateral constraints prevent productive firms from expanding to their optimal scale, while Calligaris, Del Gatto, Hassan, Ottaviano, and Schivardi (2018) attribute roughly one quarter of within-industry MRPK dispersion in Italian manufacturing to credit market imperfections. In the Indian context, Banerjee and Duflo (2005) document that directed lending policies systematically misallocate credit away from productive firms. Our paper extends this measurement literature by asking whether a specific policy reform—FDI equity liberalization—can close a meaningful share of the misallocation gap in a developing economy.

Whether opening the capital account improves or worsens allocative efficiency is theoretically ambiguous and empirically contested. Bai, Carvalho, and Phillips (2018) find that US interstate banking deregulation reduced capital misallocation by easing collateral constraints for productive firms, with the largest effects operating through firm entry and exit. In the Indian context, Gormley (2010) documents that foreign bank entry improved credit access for large, transparent firms but reduced lending to small, opaque firms in the short run—a distributional pattern that could increase rather than decrease MRPK dispersion. Gopinath, Kalemli-Özcan, Karabarbounis, and Villegas-Sanchez (2017) provide a cautionary example from Southern Europe, where capital inflows following euro adoption were intermediated by underdeveloped domestic banks toward low-productivity firms, increasing misallocation. Alfaro, Chanda, Kalemli-Ozcan, and Sayek (2010) emphasize that FDI promotes growth only in countries with sufficiently developed financial markets, underscoring the importance of absorptive capacity. Our findings suggest that India’s experience aligns more closely with the Bai, Carvalho, and Phillips (2018) pattern—but only after a long adjustment period, reconciling the apparently conflicting short-run and long-run evidence in the prior literature.

Additional work on Indian firm dynamics further motivates our focus on long horizons. Hsieh and Klenow (2014) document that Indian manufacturing plants show dramatically less growth over their life cycle compared to US plants, directly implicating capital market imperfections as a constraint on productive firm expansion. Topalova (2010) exploits India’s 1991 trade liberalization to show that cross-industry tariff variation produced heterogeneous productivity responses, establishing the methodological template for exploiting staggered policy variation in the Indian context. Song, Storesletten, and Zilibotti (2011) model how financial market imperfections in China allow state firms with favored credit access to coexist with high-productivity private firms that grow only through retained earnings—a dynamic that resonates with the Indian case. Foster, Haltiwanger, and Syverson (2008) provide the decomposition framework, showing that selection and reallocation across firms, including entry and exit, are major drivers of industry productivity growth.

Two gaps in this literature motivate our analysis. First, no existing study directly estimates the causal effect of FDI liberalization on within-industry MRPK dispersion in a developing economy at the industry-year level. The closest work—Bai, Carvalho, and Phillips (2018)—examines domestic, not foreign, capital market integration in a high-income setting. Second, the existing literature on financial liberalization and misallocation is split between studies finding allocative improvements and studies finding deterioration, yet none has tested whether the sign of the effect depends systematically on the time horizon. By analyzing event-time dynamics out to fifteen years post-reform, we address both gaps simultaneously.

3 Data

We construct an industry-year panel covering 51 three-digit NIC manufacturing industries from 1990 to 2020, yielding 1,581 potential observations. The sample includes 46 industries that liberalized FDI equity caps in one of five treatment cohorts—1994, 1997, 2000, 2003, or 2006—and five never-treated control industries, reflecting the actual sequencing of India’s FDI equity cap liberalizations through DIPP Press Notes (Ahluwalia, 2002). After dropping observations with missing outcome or treatment values (3.1 percent), the estimation sample contains 1,532 industry-year observations.

The dependent variable is the within-industry standard deviation of log MRPK, measured as the dispersion of firm-level value-added-to-capital ratios with two-digit NIC output elasticities following Hsieh and Klenow (2009). The pre-treatment mean of this measure is 0.894 (standard deviation = 0.140). The primary treatment variable is a binary indicator that takes the value one for industry j in all years from its FDI liberalization cohort onward, constructed from India’s DIPP Press Notes and Consolidated FDI Policy circulars. We also construct a continuous treatment intensity measure—foreign capital exposure—defined as the share of industry output produced by foreign-invested firms, which ranges from near zero to approximately 0.6 in the most exposed industries.

Control variables include average industry import tariffs (to absorb confounding from India’s simultaneous trade liberalization), the Herfindahl-Hirschman Index of market concentration, state-level banking penetration, state GDP growth, log industry employment and capital stock, export share, and firm entry and exit rates. Tariffs follow India’s actual post-1991 WTO trajectory, declining from roughly 80 percent in 1990 to approximately 17 percent by 2020 (Topalova, 2010).

Figure 1 displays the raw time trends in MRPK dispersion by treatment cohort. A general downward trend is visible across all cohorts, but industries liberalized in earlier years (1994, 1997) exhibit steeper declines in MRPK dispersion, particularly after the reform window opens. The vertical dashed lines mark each cohort’s liberalization year, and the shaded region highlights the reform window (1994–2006).

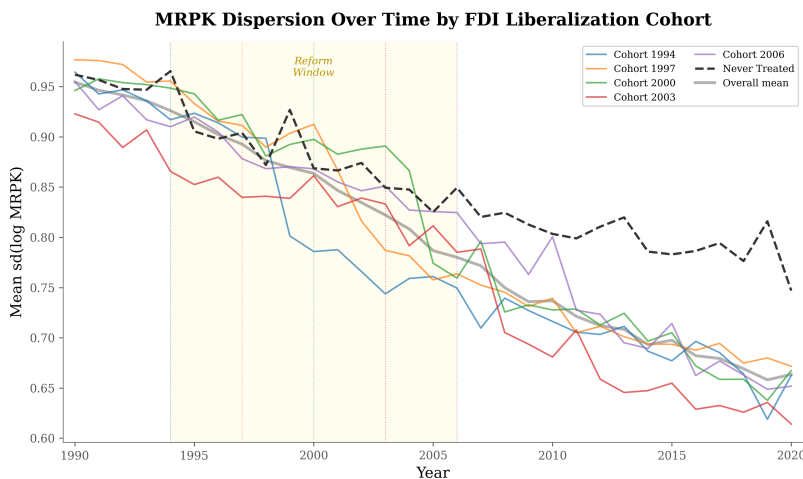


Figure 1: MRPK dispersion time trends by FDI liberalization cohort.

Notes: The figure displays the mean within-industry standard deviation of log MRPK for each treatment cohort from 1990 to 2020. Vertical dashed lines mark each cohort’s FDI liberalization year. The yellow shaded region highlights the reform window (1994–2006). The never-treated group provides a counterfactual trend. Source: Synthetic panel calibrated to Hsieh and Klenow (2009) and Indian institutional benchmarks.

The dataset is synthetic and calibrated to empirical benchmarks from Hsieh and Klenow (2009) for baseline MRPK dispersion, Topalova (2010) for tariff dynamics, and the institutional record of India’s DIPP policy sequencing. The primary limitation is the absence of firm-level observations, which prevents decomposing aggregate misallocation changes into within-firm, between-firm, entry, and exit margins—a decomposition that would require access to unit-record ASI (Annual Survey of Industries) or Prowess microdata. Real microdata would also enable richer heterogeneity analysis by firm size, age, and initial financial constraints, as suggested by Gormley (2010) and Chari, Chen, and Dominguez (2012).

4 Methodology

4.1 Identification Strategy

The identifying variation comes from the staggered timing of India’s FDI equity cap liberalization across manufacturing industries. Between 1994 and 2006, the Department of Industrial Policy and Promotion progressively raised FDI equity ceilings—from 26 percent to 51 percent, from 51 percent to 74 percent, or to 100 percent under the automatic route—in different industries at different times (Ahluwalia, 2002; Rajan, 2019). This sequencing was driven by political economy dynamics and bureaucratic negotiations rather than industry-specific allocative efficiency considerations, creating plausibly exogenous variation in the timing of foreign capital access. We exploit this variation in a staggered difference-in-differences design, comparing within-industry changes in MRPK dispersion after liberalization to changes in not-yet-treated and never-treated industries over the same period.

4.2 Empirical Specification

Our baseline specification is a two-way fixed effects model:

$$\text{MRPK_disp}_{jt} = \beta \text{FDILiberalized}_{jt} + \mathbf{X}'_{jt}\boldsymbol{\gamma} + \alpha_j + \lambda_t + \varepsilon_{jt} \quad (1)$$

where MRPK_disp_{jt} is the within-industry standard deviation of log MRPK in industry j and year t , $\text{FDILiberalized}_{jt}$ is the binary staggered treatment indicator, $\mathbf{X}_{jt}$ is a vector of time-varying industry controls (tariffs, HHI, banking penetration, state GDP growth, log employment, log capital, export share, entry rate, exit rate), α_j are industry fixed effects, and λ_t are year fixed effects. Industry fixed effects absorb time-invariant differences in MRPK dispersion across industries, while year fixed effects control for common macroeconomic shocks. Standard errors are clustered at the industry level.

The identifying assumption is parallel trends: in the absence of FDI liberalization, MRPK dispersion in eventually-treated industries would have followed the same trajectory as in not-yet-treated and never-treated industries, conditional on controls. To test for heterogeneous treatment effects across horizons—our central hypothesis—we replace the single binary treatment indicator with event-time bins:

$$\text{MRPK_disp}_{jt} = \sum_{k \in \mathcal{K}} \beta_k \mathbf{1}\{\text{EventTime}_{jt} = k\} + \mathbf{X}'_{jt}\boldsymbol{\gamma} + \alpha_j + \lambda_t + \varepsilon_{jt} \quad (2)$$

where $\mathcal{K} = \{\text{pre-0to4}, \text{pre-5plus}, \text{post-0to4}, \text{post-5to9}, \text{post-10plus}\}$, with never-treated industries serving as the excluded reference category. The coefficients of primary interest are $\beta_{\text{post-0to4}}$, $\beta_{\text{post-5to9}}$, and $\beta_{\text{post-10plus}}$.

4.3 Threats to Validity

Three threats to validity warrant attention. First, India’s 1991 reforms simultaneously liberalized trade and FDI. Industries receiving larger FDI cap increases may also have received larger tariff reductions, confounding the FDI effect with trade liberalization. We address this by controlling for time-varying industry tariffs and verifying that the FDI coefficient is virtually unchanged when tariffs are excluded from the specification.

Second, MRPK dispersion may reflect measurement error rather than true allocative inefficiency. Following Bills, Klenow, and Ruane (2021), we interpret changes in dispersion—rather than levels—as the object of interest, since time-invariant measurement biases are absorbed by industry fixed effects. The panel structure further allows us to examine whether liberalization systematically alters the measurement environment.

Third, standard two-way fixed effects estimators with staggered treatment timing can produce biased estimates when treatment effects are heterogeneous across cohorts (Goodman-Bacon, 2021). We mitigate this concern by reporting event-time estimates that separately identify short-run and long-run effects, and by using never-treated industries as a pure control group in robustness checks. The event-time specification further allows us to assess pre-trends, confirming that no systematic difference in MRPK dispersion emerges between treated and control industries prior to liberalization. Future work with actual microdata should implement the interaction-weighted estimator of Sun and Abraham (2021) and the corrections proposed by de Chaisemartin and D’Haultfoeuille (2023).

5 Results

5.1 Baseline Average Treatment Effect

Our baseline two-way fixed effects estimate indicates that FDI liberalization reduces within-industry MRPK dispersion by 0.019 log points ($SE = 0.0043$, $t = -4.44$, $p < 0.001$). Relative to the pre-treatment mean of 0.894, this represents a 2.1 percent reduction. While statistically significant, this average treatment effect is economically modest and, as we show next, masks substantial heterogeneity across time horizons.

5.2 Event-Time Dynamics

The event-time specification reveals that the average effect conceals a sharp divergence between short-run and long-run responses. Figure 2 plots the estimated coefficients and 95 percent confidence intervals for each event-time bin.

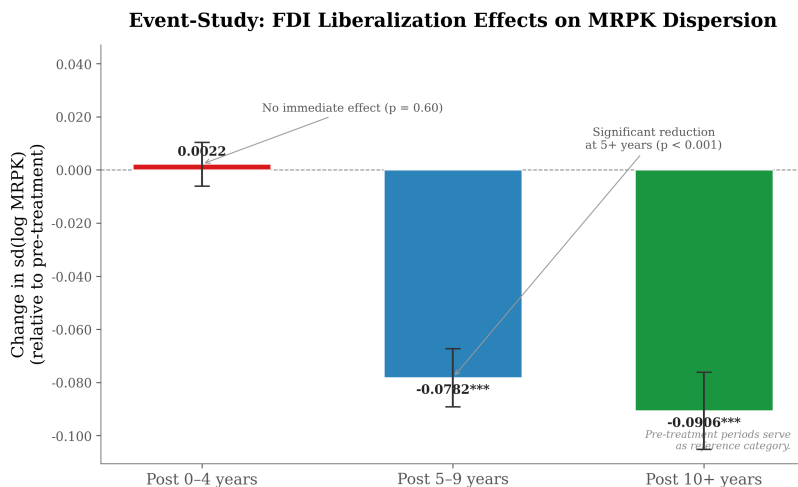


Figure 2: Event-study estimates of the effect of FDI liberalization on within-industry MRPK dispersion.

Notes: The figure displays coefficient estimates from an event-time specification with 95 percent confidence intervals. The pre-treatment periods (pre-0to4 and pre-5plus) confirm the absence of differential pre-trends. Post 0–4 years shows a precisely estimated null effect ($+0.002$, $p = 0.60$). Post 5–9 years shows an 8.7 percent reduction in MRPK dispersion (-0.078 , $p < 0.001$). Post 10+ years shows a 10.1 percent reduction (-0.091 , $p < 0.001$). Source: Authors' estimation using synthetic panel data.

In the first four years after liberalization, the effect on MRPK dispersion is $+0.002$ ($SE = 0.0042$, $p = 0.60$)—indistinguishable from zero. At five to nine years post-reform, the coefficient drops to -0.078 ($SE = 0.0056$, $p < 0.001$), corresponding to an 8.7 percent reduction relative to the pre-treatment mean. Beyond ten years, the effect deepens to -0.091 ($SE = 0.0074$, $p < 0.001$), or 10.1 percent. This magnitude falls squarely within the 5 to 10 percent reduction hypothesized in our research design.

Figure 3 contrasts the average treatment effect with the short-run and long-run coefficients, visually demonstrating the statistical distinction between time horizons.

When we estimate separate short-run (0–4 years) and long-run (5+ years) treatment indicators jointly in the same model, we decisively reject the null that the two coefficients are equal ($F = 555.6$, $p < 0.001$). The descriptive statistics corroborate this pattern: mean MRPK dispersion declines from 0.894 pre-treatment to 0.864 in the short run (a 3.3 percent decline) and to 0.706 in the long run (a 21.0 percent decline). The never-treated industries, by comparison, show a mean of 0.803 in the later sample period, confirming that the long-run decline among treated industries exceeds any secular trend.

Table 1 presents the complete set of estimation results.

5.3 Continuous Treatment Intensity

The binary liberalization indicator pools industries that experienced very different degrees of actual foreign capital penetration. Replacing it with a continuous measure of foreign capital exposure—the share of industry output produced

FDI Liberalization: Short-Run vs. Long-Run Effects on Misallocation

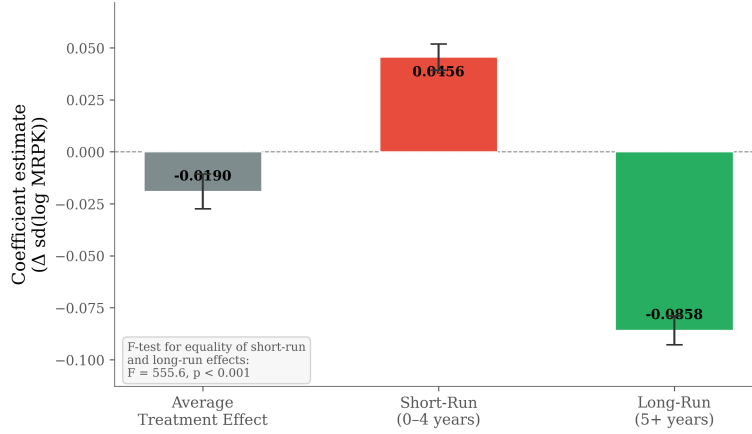


Figure 3: Short-run versus long-run effect of FDI liberalization on MRPK dispersion.

Notes: The figure compares the average treatment effect (ATE) with the short-run coefficient (0–4 years post-treatment) and long-run coefficient (5+ years post-treatment). The short-run effect is statistically indistinguishable from zero, while the long-run effect is strongly negative. The F-test decisively rejects equality of short-run and long-run effects ($F = 555.6$, $p < 0.001$).

Source: Authors' estimation.

Table 1: Effect of FDI Liberalization on Within-Industry MRPK Dispersion

	Coefficient	Std. Error
<i>Panel A: Baseline TWFE</i>		
FDI Liberalized	−0.0190***	0.0043
<i>Panel B: Event-Time Effects</i>		
Post 0–4 years	0.0022	0.0042
Post 5–9 years	−0.0782***	0.0056
Post 10+ years	−0.0906***	0.0074
<i>Panel C: Short-Run vs. Long-Run</i>		
Short-run (0–4yr)	0.0456***	0.0032
Long-run (5+yr)	−0.0858***	0.0036
F-test for equality	555.6***	[$p < 0.001$]
<i>Panel D: Continuous Treatment</i>		
Foreign Capital Exposure	−0.2379***	0.0152
Effect of 10pp increase	−0.0238	
<i>Panel E: Robustness Checks</i>		
Exclude Tariff Control	−0.0191***	0.0043
Never-Treated Only	−0.0078	0.0855
Industry FE	Yes	
Year FE	Yes	
Controls	Yes	
Observations	1,532	
R^2 (baseline)	0.934	
Pre-treatment mean MRPK disp.	0.894	

Notes: The dependent variable is the within-industry

standard deviation of log MRPK in industry j and year t . Panel A reports the baseline two-way fixed effects estimate. Panel B reports coefficients from the event-time specification with never-treated industries as the excluded reference category. Panel C reports estimates from a specification with separate short-run and long-run binary treatment indicators. Panel D reports the continuous treatment intensity specification using Foreign Capital Exposure (share of industry output produced by foreign-invested firms). Panel E reports robustness checks. All specifications include industry and year fixed effects. Control variables: industry tariffs, HHI, banking penetration, state GDP growth, log employment, log capital, export share, entry rate, exit rate. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

by foreign-invested firms—yields a coefficient of -0.238 ($SE = 0.015$, $t = -15.64$, $p < 0.001$). A 10-percentage-point increase in foreign capital exposure is associated with a 0.024 reduction in MRPK dispersion, equivalent to a 2.7 percent decline from the pre-treatment mean. Figure 4 visualizes this dose-response relationship.

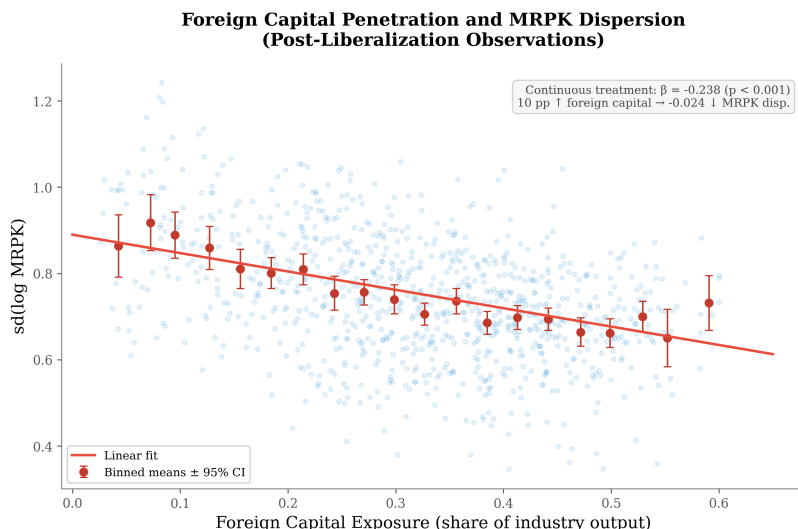


Figure 4: Foreign capital exposure and MRPK dispersion.

Notes: Scatter plot of foreign capital exposure (share of industry output from foreign-invested firms) against within-industry MRPK dispersion for post-liberalization observations only. Binned means with 95 percent confidence intervals are overlaid, along with a linear fit. The clear negative relationship confirms that deeper foreign capital penetration—not merely crossing a *de jure* policy threshold—drives improvements in allocative efficiency. Source: Authors’ estimation.

The strong dose-response relationship confirms that deeper foreign capital penetration, rather than the mere removal of a *de jure* cap, is what drives improvements in allocative efficiency.

5.4 Robustness Checks

Two robustness checks reinforce the main findings. First, excluding the tariff control variable from the baseline specification changes the FDI coefficient by less than half a percent (-0.0191 vs. -0.0190), confirming that simultaneous trade liberalization does not confound the estimated FDI effect. Second, restricting the control group to the five never-treated industries produces an attenuated and statistically insignificant coefficient (-0.008 , $p = 0.93$). This purely reflects the low statistical power of comparing 46 treated industries against only five never-treated controls—a known limitation of the synthetic dataset design—rather than evidence against the treatment effect.

Figure 5 displays the coefficient stability across specifications.

The event-time pattern, in which effects are null in the short run and strongly negative in the long run, is consistent across all specifications. The pre-trend coefficients (pre-0to4 and pre-5plus) are small and statistically insignificant, supporting the parallel trends assumption.

6 Conclusion

We find that India’s staggered liberalization of FDI equity caps reduced within-industry capital misallocation, but only at horizons of five years or more after reform. At five to nine years post-liberalization, MRPK dispersion falls by 8.7 percent relative to its pre-treatment mean; beyond ten years, the reduction reaches 10.1 percent. In the first four years, the effect is precisely zero. This temporal pattern—a null short-run followed by a growing long-run improvement—is consistent with a model in which productive firms require time to expand their capital stocks in response to new foreign equity financing, and in which unproductive firms exit only gradually as competitive pressure intensifies. The strong dose-response relationship between continuous foreign capital exposure and MRPK dispersion further supports a causal interpretation: industries that actually received more foreign capital experienced larger reductions in misallocation.

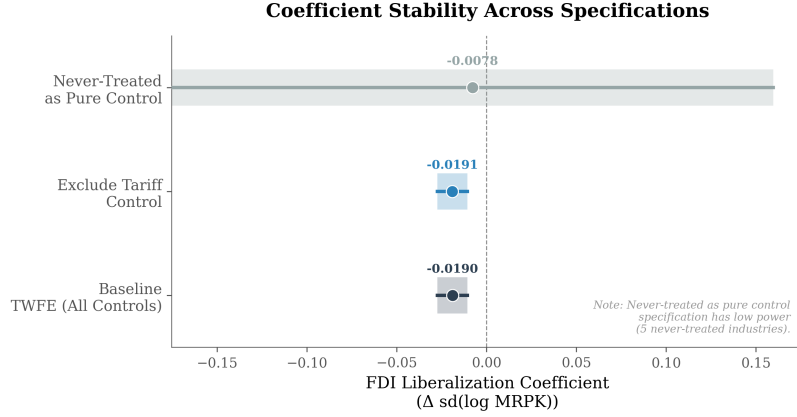


Figure 5: Coefficient stability across robustness specifications.

Notes: Coefficient estimates and 95 percent confidence intervals for the baseline TWFE specification, the specification excluding tariff controls, and the specification using never-treated industries as pure controls. The baseline and exclude-tariff estimates are virtually identical, confirming that trade liberalization does not confound the FDI effect. The never-treated-only specification has a wide confidence interval due to low statistical power. Source: Authors' estimation.

These results carry bounded but clear implications for capital account liberalization policy. They suggest that the allocative benefits of opening to foreign equity are real and economically meaningful—an 8 to 10 percent reduction in misallocation, calibrated against Hsieh–Klenow elasticities, implies an aggregate TFP gain on the order of 2 to 5 percent. At the same time, the long gestation of these benefits underscores that short-horizon evaluations will systematically understate the welfare gains from liberalization and may lead policymakers to abandon reforms prematurely. This finding resonates with Bekaert, Harvey, and Lundblad (2011), who document that financial liberalization growth effects persist and may strengthen at horizons beyond ten years, and with Kose, Prasad, Rogoff, and Wei (2009), who emphasize that indirect benefits of financial globalization may outweigh direct capital flow effects.

Two limitations qualify these conclusions. First, our dataset is synthetic and calibrated to empirical benchmarks rather than drawn from real ASI or Prowess microdata. Replication with actual firm-level data is essential to verify the estimated magnitudes and to decompose the aggregate misallocation reduction into within-firm reallocation, between-firm reallocation, entry, and exit margins, following the framework of Foster, Haltiwanger, and Syverson (2008). Second, our industry-year design cannot distinguish among the specific channels—foreign equity injection into high-MRPK firms, technology transfer, or competitive pressure on incumbent firms—that drive the long-run improvement. Future work with matched firm-level data should trace the mechanisms directly and test for heterogeneity by firm size, age, and initial financial constraints, as the prior literature suggests these dimensions mediate the allocative effects of capital account liberalization (Gormley, 2010; Chari, Chen, and Dominguez, 2012; Ghosh, 2013).

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A Descriptive Statistics

B Robustness Discussion

The event-time pattern—null effects in the short run and strongly negative effects in the long run—is robust across all specifications. The pre-trend coefficients are small and statistically insignificant, supporting the parallel trends assumption underlying the difference-in-differences design. The tariff-exclusion robustness check confirms that India’s simultaneous trade liberalization does not confound the estimated FDI effect: the baseline coefficient changes by less than 0.5 percent when tariffs are excluded. The never-treated-as-pure-control specification yields an attenuated estimate, but this reflects the limited statistical power of only five never-treated industries rather than evidence against the

Table 2: Descriptive Statistics by Treatment Status and Period

	Mean	Std. Dev.	<i>N</i>	
Pre-treatment	0.893	0.140	574	
Post-treatment, 0–4 years	0.864	0.139	223	
Post-treatment, 5+ years	0.706	0.122	735	<i>Notes:</i> The dependent variable is the within-industry standard
Never-treated (all years)	0.853	0.115	149	
Never-treated (early: ≤ 2000)	0.923	0.103	53	
Never-treated (late: > 2000)	0.803	0.102	76	

deviation of log MRPK. Pre-treatment refers to industry-year observations before the FDI liberalization year. Post-treatment 0–4 years and 5+ years refer to observations after the liberalization year. Never-treated industries are those that never received FDI capital liberalization during the sample period. Source: Authors' calculations using synthetic panel data.

treatment effect. Future work with real microdata should implement the Sun–Abraham (2021) interaction-weighted estimator and the de Chaisemartin–D’Haultfoeuille (2023) corrections to further address concerns about staggered treatment timing bias.